Delivery Metrics at a Glance

What each delivery metric measures, the question it answers, and how it gets misused.

Manage Flow · Draws on Kanban flow metrics (Daniel Vacanti), the DORA research and Little's Law

TEAM _____

DATE _____

Velocity

MEASURES
Story points a team finishes in an iteration.

ASKS
"How much can we plan into the next iteration?"

MISUSED WHEN
Set as a target, or compared across teams.

IN THIS KIT **Estimation**

Throughput

MEASURES
Items finished per week or iteration, counted.

ASKS
"How many items can we expect next month?"

MISUSED WHEN
Counting items of very different sizes.

IN THIS KIT **Release Forecast & Burn-up**

Cycle time

MEASURES
Time from starting an item to finishing it.

ASKS
"Once we start, how long does it take?"

MISUSED WHEN
Averages hide the slow ones: use percentiles.

IN THIS KIT **Ageing WIP & SLE**

Lead time

MEASURES
Time from request to delivery to the customer.

ASKS
"How long does the customer wait?"

MISUSED WHEN
Not saying where the clock starts.

IN THIS KIT **Value Stream Map**

Work item age

MEASURES
How long an unfinished item has been in progress.

ASKS
"Which items are stuck right now?"

MISUSED WHEN
Only looking once items finish.

IN THIS KIT **Ageing WIP & SLE**

Work in progress

MEASURES
Items started but not yet finished.

ASKS
"Are we starting more than we finish?"

MISUSED WHEN
Raising the limit whenever it's hit.

IN THIS KIT **Kanban Board Design**

Flow efficiency

MEASURES
Share of lead time spent working, not waiting.

ASKS
"How much of the time is waiting?"

MISUSED WHEN
Making people busier, not cutting waits.

IN THIS KIT **Value Stream Map**

Cumulative flow

MEASURES
Items in each state over time, as bands.

ASKS
"Where is work piling up?"

MISUSED WHEN
A widening band is a queue: act on it.

IN THIS KIT **Kanban Cadences**

Burn-up and burndown

MEASURES
Work done, and total scope, over time.

ASKS
"Will we finish by the date?"

MISUSED WHEN
A burndown hides added scope.

IN THIS KIT **Release Forecast & Burn-up**

DORA four

MEASURES
Deploy frequency, lead time for changes, failure rate, time to restore.

ASKS
"Are we delivering faster and more safely?"

MISUSED WHEN
Chasing speed and ignoring stability.

IN THIS KIT **Delivery Metrics & Tech Debt**

Little's Law

WHY LIMITING WIP SPEEDS THINGS UP

Cycle time = WIP / Throughput

(average items in progress / average items finished per week)

20 items in progress / 5 finished a week
= about 4 weeks for each item.

Halve the work in progress to 10, at the same throughput, and items take about 2 weeks.

Holds for averages over a stable period. From queueing theory (John Little, 1961).

Which Metric for Which Question?

START WITH THE QUESTION

When will it be done?	throughput, as a forecast range
What's stuck right now?	work item age
Why is it slow?	flow efficiency, cumulative flow
What can we promise?	85th percentile cycle time
How much fits next?	velocity or throughput
Are releases safe?	change failure rate, time to restore

Use Metrics Well

GOODHART'S LAW APPLIES

Teams, never individuals
Personal metrics get gamed and hurt teamwork.

Ranges, not single numbers
"13–20 June" is honest; "15 June" isn't.

Trends, not snapshots
One iteration tells you little; six tell a story.

Speed with quality
Pair every speed measure with a quality one.

Questions, not targets
When a measure becomes a target, it stops measuring.

"Which metric would tell us most about our biggest problem right now?"

"Which of our measures has quietly become a target?"

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